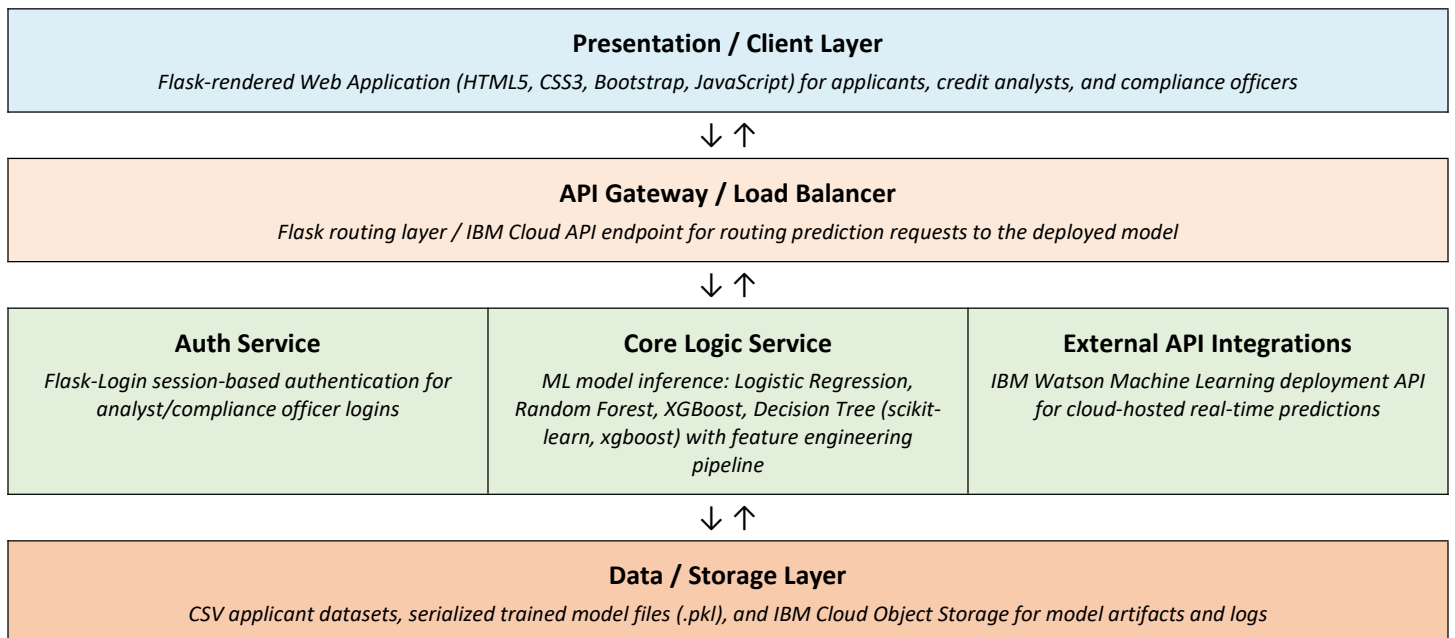


Solution Architecture

Date	02 JULY 2026
Team ID	CCAP-01
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the web interface through which analysts, compliance officers, and applicants enter financial/demographic details and view instant approval or rejection predictions.	HTML5, CSS3, Bootstrap, JavaScript, Flask templates (Jinja2)
API Gateway	Routes incoming prediction requests from the client to the appropriate backend service and forwards responses from the model service back to the UI.	Flask routing, IBM Cloud API Gateway
Microservices	Core logic service that performs feature engineering and runs the trained classification model to predict credit card approval/rejection in real time.	Python, scikit-learn, XGBoost, pandas, numpy, IBM Watson Machine Learning
Database	Stores historical applicant data used for training, the serialized trained model files, and prediction logs for auditing and compliance review.	CSV datasets, Pickle (.pkl) model files, IBM Cloud Object Storage

